1 I

1.1 (1)

Since the electromagnetic wave is a plane wave propagating in the positive x direction, we have

$$\vec{E} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \vec{B} \times \hat{k}, \qquad \hat{k} = \hat{x}.$$

With $\vec{B} = A \sin(kx - \omega t)\hat{y}$, the electric field is

$$\boxed{\vec{E} = -\frac{A}{\sqrt{\mu_0 \epsilon_0}} \sin(kx - \omega t)\hat{z}}.$$

1.2 (2)

The Poynting vector is

$$\begin{aligned} \vec{S} &= \frac{1}{\mu_0} \vec{E} \times \vec{B} \\ &= \frac{A^2}{\mu_0 \sqrt{\mu_0 \epsilon_0}} \sin^2(kx - \omega t)\hat{x} \\ &= \sqrt{\frac{\epsilon_0}{\mu_0}} \frac{A^2}{\mu_0 \epsilon_0} \sin^2(kx - \omega t)\hat{x}. \end{aligned}$$

Hence,

$$\boxed{|\vec{S}| = \sqrt{\frac{\epsilon_0}{\mu_0}} |\vec{E}|^2}.$$

2 II

2.1 (3)

The component of the electric field parallel to the interface must be continuous. Using the directions defined in the figure,

$$E_1 \cos\alpha - E_1' \cos\alpha = E_2 \cos\beta.$$

Therefore,

$$\boxed{E_2 = \frac{\cos\alpha}{\cos\beta}(E_1 - E_1')}.$$

2.2 (4)

The component of the magnetic field strength $\vec{H} = \vec{B}/\mu$ parallel to the interface is also continuous:

$$\frac{B_1 + B_1'}{\mu_1} = \frac{B_2}{\mu_2}.$$

Thus,

$$\boxed{B_2 = \frac{\mu_2}{\mu_1}(B_1 + B_1')}.$$

2.3 (5)

For a plane electromagnetic wave in a medium, the electric and magnetic field amplitudes satisfy

$$\frac{E_i}{B_i} = \frac{1}{\sqrt{\epsilon_i \mu_i}}, \qquad \frac{E_i}{B_i/\mu_i} = Z_i = \sqrt{\frac{\mu_i}{\epsilon_i}}.$$

Therefore, the boundary condition from (4) becomes

$$\frac{E_2}{Z_2} = \frac{E_1 + E_1'}{Z_1}.$$

Combining this with (3),

$$\frac{\cos\alpha}{\cos\beta}(E_1 - E_1') = \frac{Z_2}{Z_1}(E_1 + E_1').$$

Solving for the reflection and transmission amplitude ratios gives

$$\boxed{\frac{E_1'}{E_1} = \frac{\dfrac{\cos\alpha}{\cos\beta} - \dfrac{Z_2}{Z_1}}{\dfrac{\cos\alpha}{\cos\beta} + \dfrac{Z_2}{Z_1}}},$$

and

$$\boxed{\frac{E_2}{E_1} = \frac{2}{\dfrac{\cos\beta}{\cos\alpha} + \dfrac{Z_1}{Z_2}}}.$$

2.4 (6)

When $\mu_1 = \mu_2$, the relative refractive index is

$$n = \frac{Z_1}{Z_2} = \frac{\sin\alpha}{\sin\beta}.$$

For $\alpha = 45°$ and $\beta = 30°$,

$$n = \frac{\sin 45°}{\sin 30°} = \sqrt{2}.$$

Substituting these values into the expressions from (5),

$$\frac{E_2}{E_1} = \frac{2}{\dfrac{\cos 30°}{\cos 45°} + \sqrt{2}} = \frac{2}{\sqrt{3/2} + \sqrt{2}} \approx \boxed{0.8},$$

$$\frac{E_1'}{E_1} = \frac{\dfrac{\cos 45°}{\cos 30°} - \dfrac{1}{\sqrt{2}}}{\dfrac{\cos 45°}{\cos 30°} + \dfrac{1}{\sqrt{2}}} \approx \boxed{0.07}.$$

Both results are given to one significant figure.

2.5 (7)

Using $Z_2/Z_1 = \sin\beta/\sin\alpha$ in the reflection coefficient from (5), we obtain

$$\begin{aligned}\frac{E_1'}{E_1} &= \frac{\sin\alpha\cos\alpha - \sin\beta\cos\beta}{\sin\alpha\cos\alpha + \sin\beta\cos\beta} \\ &= \frac{\sin(\alpha-\beta)\cos(\alpha+\beta)}{\sin(\alpha+\beta)\cos(\alpha-\beta)}.\end{aligned}$$

For the given case $\alpha > \beta$, with $0 \le \alpha, \beta < 90°$, the nontrivial condition for vanishing reflection is

$$\cos(\alpha+\beta) = 0.$$

Hence,

$$\boxed{\alpha + \beta = 90°}.$$

This is Brewster's condition for p-polarized light. The corresponding incidence angle α_B is called *Brewster's angle*. Since $n = \sin\alpha/\sin\beta$,

$$\boxed{\tan\alpha_B = n}.$$

3 III

3.1 (8)

Natural sunlight is generally unpolarized and contains two polarization components: p-polarized light, whose electric field oscillates parallel to the plane of incidence, and s-polarized light, whose electric field oscillates perpendicular to that plane.

At Brewster's angle, the reflection coefficient for p-polarized light vanishes. Consequently, reflected sunlight from the water surface is dominated by the s-polarized component near this angle. A polarizing filter that blocks this component suppresses surface glare while transmitting more of the light coming from underwater, making objects beneath the surface easier to observe.